Otopair Enrichment Pipeline

v9 Roadmap | Week 1 Summary | April 2, 2026

Executive Summary

Week 1 focused on hardening the enrichment pipeline, connecting labor data to the booking flow, fixing data accuracy issues, and adding resilience to the batch processing system. Of the 19 tasks in the v9 roadmap, **11 are complete**, 2 are blocked on external dependencies, and 6 remain for Weeks 2-3.

Tasks Completed	**11 / 19**
Completion Rate	**58%**
Blocked	**2 (external deps)**
Remaining	**6 (Weeks 2-3)**

Full Task Status

#	Task	Status	Week
1	NHTSA VIN Decode Integration	**DONE**	Wk 1
2	VDB Advanced VIN Decode V2	**DONE**	Wk 1
3	Batch API Pipeline (Tier 2)	**DONE**	Wk 1
4	Normalized Schema + Write Layer	**DONE**	Wk 1
5	Quotable Labor Wiring (SVS Bridge)	**DONE**	Wk 1
5A	VDB Repair Estimates API + Labor Mapping	**DONE**	Wk 1
8	Drivetrain Consistency (FWD fluid nulling)	**DONE**	Wk 1
9	is_staggered Auto-Computation	**DONE**	Wk 1
11	Fill Rate Accuracy Fix	**DONE**	Wk 1
12	Tier 2 Resilience (3 fixes)	**DONE**	Wk 1
13	Source Registry Logging	**DONE**	Wk 1
6	Oto AI Scope Definition	**BLOCKED**	Wk 2
14	Mechanic UI Integration	**BLOCKED**	Wk 2
7	Evidence + Consensus System	PENDING	Wk 2
10	Empirical Learning Expansion	PENDING	Wk 2
15	Monthly Price Refresh Cron	PENDING	Wk 2
16	Admin Dashboard for Pipeline Health	PENDING	Wk 3
17	Content Sanitization	PENDING	Wk 3
18	Legacy File Cleanup	PENDING	Wk 3
19	Handoff Document Update	PENDING	Wk 3

Week 1 Completed Work — Details

Task 1-4: Core Pipeline Build

- NHTSA VIN decode with full VPIC data extraction
- VDB Advanced VIN Decode V2 integration with local cache + fallback
- Tier 2 Batch API pipeline: Batch 1A (fluids/intervals) + 1B (parts/trim) + Batch 2 (gap fill)
- Normalized schema across engines, transmissions, drivetrain_configs, trim_specs, OEM parts, service_intervals, labor_times, part_prices, and enrichment_evidence

Task 5: Quotable Labor Wiring

- Bridged v3 labor_times table with legacy service_vehicle_specs used by the booking/pricing flow
- Priority chain: empirical (0.95 confidence, 3+ samples) > VDB repair (0.90) > web_search (0.85) > training_data (0.75)
- Added empirical feedback loop in job_actuals.ts: each completed job patches labor_times with running average
- Frontend booking flow now automatically gets VDB and enrichment-sourced labor hours without code changes

Task 5A: VDB Repair Estimates API

- Resolved correct endpoint: /repair-estimates/{vin} (after 5 iterations)
- Flattened VDB's nested mileage block structure to extract labor type + hours only
- Built VDB_TO_SERVICE_SLUG mapping (25+ VDB strings to Otopair service slugs)
- Removed all mileage interval references per product decision (enrichment handles intervals)

Task 8: Drivetrain Consistency

- FWD vehicles now correctly null out diff_fluid_type and tc_fluid_type
- AWD/4WD sets has_transfer_case=true; RWD/FWD sets has_transfer_case=false
- Applied to both Batch 1 resolved write and NHTSA fallback paths

Task 9: is_staggered Auto-Computation

- Auto-computed from front vs rear tire size comparison before upsertTrimSpecs
- No AI involvement needed; pure string comparison

Task 11: Fill Rate Accuracy

- Expanded trim fields from 7 to 9 (added battery_type, battery_location)
- FWD drivetrain no longer inflates fill rate by counting absent fluid fields
- Service intervals and labor use applicableServices.length as denominator

Task 12: Tier 2 Resilience (3 Fixes)

- Fix 1: Concurrent run prevention with 4-hour safety valve for stuck runs
- Fix 2: Batch 1 + Batch 2 submitBatch calls wrapped in try-catch with run failure cleanup
- Fix 3: Errored/expired batch requests now detected and surfaced instead of silently returning empty data

Task 13: Source Registry Logging

- Unregistered domains now logged with accuracy stats instead of silently skipped
- Enables manual or automated addition to source_registry for new data sources

Key Bug Fix: VDB Placeholder Engine Codes

VDB returns placeholder engine codes like STDEN, STD, STDE for vehicles it cannot identify. The pipeline was treating these as real codes, skipping the entire fallback chain.

Added a blocklist of 9 known placeholder codes. When detected, the pipeline now correctly falls back through NHTSA engine extraction, Claude AI normalization, web search + Haiku verification, and synthetic code generation. Tested with a 2022 Toyota RAV4 Prime SE (VIN: JTMCB3FV4ND089342) which successfully resolved to A25A-FXS.

Blocked Tasks

Task 6: Oto AI

Needs Waleed's scope definition for AI-powered vehicle insights feature.

Task 14: Mechanic UI

Depends on Daniel's frontend work for mechanic-facing dashboard integration.

Week 2-3 Outlook

Week 2: Evidence + Consensus System (Task 7), Empirical Learning Expansion (Task 10), Monthly Price Refresh Cron (Task 15)

Week 3: Admin Dashboard (Task 16), Content Sanitization (Task 17), Legacy Cleanup (Task 18), Handoff Doc (Task 19)
